$C(15+3-1, 15)$
↑
sum

HW5 due 2/16

1) 12) ABCDE DB AE C
$3! = \boxed{6}$

14) _ _ _ _ _ _ $\frac{5!}{2} = \boxed{60}$
n

16) ~~AB BE~~ COE ACD
Total − AB | BE AB BE AB+BE
$5! - 42 =$ $(4! + 4!) - 3! = 42$
$120 - 42 = \boxed{78}$ or $5! - (2 \cdot 4! - 3!)$

17) _ _ _ _ _ BD can go in any of the spots
↳ 2!
$\binom{5}{3}(2!) = \frac{5!2!}{(5-3)!3!} = \frac{5!}{3!} = 5 \cdot 4 = \boxed{20}$

2) 44) 40 48 40 48 4 cards of each rank in deck $\binom{4}{1} = 1$
$\binom{13}{1}$ 5th card: 48 cards left (52-4)

$\binom{13}{1}\binom{4}{4}\binom{48}{1} = 13 \cdot 1 \cdot 48 = \boxed{13 \cdot 48} = \boxed{624 \text{ hands}}$

45) $\boxed{\binom{13}{5}}$ 13 spade cards, choose 5 $\frac{13!}{5!8!}$

47) $\binom{4}{1}\boxed{\binom{13}{2}}(13)(13)(13) = \boxed{4 \cdot 13 \cdot 13 \cdot 13 \cdot \binom{13}{2}}$
one suit has 2

48) 4 suits, only 1 A2345 in each $4 \cdot 1 = \boxed{4}$

3) 26) $x_1 - 1 \geq 0$ $x_2 - 1 \geq 0$ $x_3 - 1 \geq 0$ $(x_1 - 1) + (x_2 - 1) + (x_3 - 1) = 12$

$\binom{12+3-1}{3-1} = \boxed{\binom{14}{2}}$

4) 33) $x_1 + x_2 + \ldots x_n + q = M$ so, $q = M - (x_1 + x_2 \ldots x_n)$
M stars, n bars $C(M+n, n)$

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$$

5) (a) $0 = \sum_{k=0}^{n} (-1)^k C(n,k)$

Using Binomial theorem for $a=1, b=1$,

$$(1-1)^n = \sum_{k=0}^{n} \binom{n}{k} 1^{n-k} (-1)^k$$

$$0 = \sum_{k=0}^{n} \binom{n}{k} (-1)^k$$

These are equal

6) 19) $C(n,k)$ is choosing k people from n

$C(n,n-k)$ is NOT choosing $n-k$ people from n, which means choosing k people from n as well.

$$C(n,k) \stackrel{?}{=} C(n,n-k)$$

$$\frac{n!}{k!(n-k)!} \stackrel{?}{=} \frac{n!}{(n-k)!(n-(n-k))!}$$

$$\frac{n!}{k!(n-k)!} \stackrel{\checkmark}{=} \frac{n!}{(n-k)!\,k!}$$

7) 28) Use induction, base step, $n=1$:

$$2C(2\cdot1-1,1) = C(2,1)$$

$$2C(1,1) = C(2,1)$$

$$2 = 2 \quad \checkmark$$

Induction step: $n+1$

$$2C(2(n+1)-1, n+1) = C(2(n+1), n+1)$$

LHS: $2C(2n+1, n+1) = 2\left(\frac{(2n+1)!}{(n+1)!(2n+1-(n+1))!}\right) = 2\left(\frac{(2n+1)!}{(n+1)!\,n!}\right) \cdot \frac{(n+1)}{(n+1)} = 2\left[\frac{(2n+1)!(n+1)}{(n+1)!(n+1)!}\right]$

$$\frac{2n+2\,(2n+1)!}{(n+1)!(n+1)!} = \left[\frac{(2n+2)!}{(n+1)!(n+1)!}\right]$$

RHS: $C(2(n+1), n+1) = C(2n+2, n+1) = \frac{(2n+2)!}{(n+1)!(2n+2-(n+1))!} = \left[\frac{(2n+2)!}{(n+1)!(n+1)!}\right]$

LHS = RHS so $2C(2n-1, n) = C(2n, n)$ for all $n \geq 1$.

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$$

8) 24) $\sum_{k=0}^{n/2} C(n, 2k) = 2^{n-1} = \sum_{k=1}^{n/2} C(n, 2k-1)$

binomial theorem

↳ Let $a, b = 1$

$$(1+1)^n = \sum_{k=0}^{n} \binom{n}{k} (1)^n (1)^0$$

$$2^n = \sum_{k=0}^{n} \binom{n}{k}$$

We can split this summation into even and odd numbers by dividing n by 2. evens start at 0 and are represented by $2k$, whereas odds start at 1, and are represented by $2k-1$ so we only get even and odd numbers, respectively.

$$\frac{2^n}{2} = 2^{n-1}$$

$$\sum_{k=0}^{n/2} C(n, 2k) = 2^{n-1} = \sum_{k=1}^{n/2} C(n, 2k-1)$$